

Deeploy: Enabling Energy-Efficient Deployment of Small Language Models On Heterogeneous Microcontrollers

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Abstract—With the rise of Embodied Foundation Models (EFMs), most notably Small Language Models (SLMs), adapting Transformers for edge applications has become a very active field of research. However, achieving end-to-end deployment of SLMs on microcontroller (MCU)-class chips without high-bandwidth off-chip main memory access is still an open challenge. In this paper, we demonstrate high-efficiency end-to-end SLM deployment on a multicore RISC-V (RV32) MCU augmented with ML instruction extensions and a hardware neural processing unit (NPU). To automate the exploration of the constrained, multi-dimensional memory vs. computation tradeoffs involved in aggressive SLM deployment on heterogeneous (multicore+NPU) resources, we introduce Deeploy, a novel Deep Neural Network (DNN) compiler, which generates highly-optimized C code requiring minimal runtime support. We demonstrate that Deeploy generates end-to-end code for executing SLMs, fully exploiting the RV32 cores’ instruction extensions and the NPU: We achieve leading-edge energy and throughput of 490 μ J/Token, at 340 Token/s for an SLM trained on the TinyStories dataset, running for the first time on an MCU-class device without external memory.

Index Terms—Neural Networks, TinyML, Embodied AI, Foundation Models, Accelerators, Compilers

I. INTRODUCTION

The latest evolutions in mainstream Artificial Intelligence (AI) have been driven by Transformers, which have taken over from Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) as the leading edge models for language processing and multi-modal applications [1], [2]. The success of Transformers can be primarily attributed to the emergence of the Foundation Model (FM) paradigm: large Transformer models extensively pre-trained on datasets spanning trillions of tokens and then fine-tuned with a much lower volume of labeled data to solve domain-specific problems. Following the success of FMs in Natural Language Processing (NLP) [1], [3], an increasing number of fields

are starting to formulate and adapt FMs for high dimensional sensor data that has traditionally been challenging to process, like decoding neural data [4], [5], or training embodied AI agents [6], [7], which may incorporate multi-modal sensor inputs.

Operating directly on sensory data and in a cyber-physical loop may lead to solving many outstanding challenges in fields such as brain-machine interfaces [5] and miniaturized robotics [7]. However, to materialize this promise, models of this class need to be *embodied* in physical devices as Embodied Foundation Models (EFMs), and they must cope with the strict constraints in terms of compute throughput, power consumption, and footprint typical of edge devices. Unlike datacenter-scale systems, which collect and aggregate sensor data over sharded resources for high-throughput processing, embodied AI systems must process sensor data with extremely low latency and memory capacity under tight power constraints. This is particularly challenging for the smallest class of AI-oriented computers: so-called “*tinyML*” devices operating at the extreme edge, based on microcontroller-class devices without complex operating systems or Memory-Management Units (MMUs), relying on user-level software to implement low-level hardware management functionalities. Despite many recent successes with previous-generation Deep Neural Networks (DNNs), the emergence of the tinyML paradigm for EFMs faces the dual challenge of reducing FMs to a manageable size and enabling their deployment on tiny devices.

A first concrete step in this direction is the recent introduction of Small Language Models (SLMs): FMs with tens to a few hundred million, rather than several billion parameters [8], [9]. While most currently available FMs are focused on processing natural language at a proof-of-concept scale, the effort towards embedded multi-modal sensor inputs with small-scale, application-specific FMs offers a highly promising path for the development of this novel class of models. Much like what happened with the initial emergence of Deep Learning [10], the evolution of advanced tinyML applications based on EFMs is currently prevented by the lack of suitable targets for deployment of these models and, even more, of deployment frameworks that enable utilizing existing specialized hardware to its full capabilities.

Deploying tiny EFMs requires overcoming several challenges specific to the tinyML domain. Large-scale AI inference systems typically employ heterogeneous computer architectures composed by a conventional host (e.g., an x86 processor) and a very large throughput-oriented accelerator (e.g., H100 [11], TPU [12]), which is fully exploited only at large batch sizes. Conversely, tinyML is used for latency-sensitive applications focusing on real-time inference without batching. As a consequence, tinyML AI inference typically employs much more specialized accelerator architectures [13],

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